

content to consumer electronic devices required significant investments by television and component manufacturers because of the need to integrate powerful intelligence and processing on the devices themselves. But now the iPoint 3D system from Fraunhofer, Germany, allows users to communicate with a 3D display through simple gestures—without touching it and without 3D glasses or a data glove.

At the heart of iPoint 3D is a recognition device—not much larger than a keyboard—that can be suspended from the ceiling above the user or integrated in a coffee table. Its two built-in cameras detect hands and fingers in real time and transmit the information to a computer. The system responds instantly, as soon as someone in front of the screen moves his hands. No physical contact or special markers are involved.

The small device is equipped with two FireWire cameras—inexpensive, off-the-shelf video cameras that are easy to install.

This form of giving commands would make many situations safer and more pleasant than they are today. Drivers, for example, could operate their car radios and navigators more easily, and TV viewers at home in their armchairs would no longer need a remote control to flick through the channels. A whole new generation of video games could be created if the technology involved were able to identify and interpret human gestures. Even machines, household appliances or videoconference systems could be controlled merely by hand signals. The system could also be of help to physically disabled people, enabling them to interact with a computer without the need for a mouse and keyboard.

Technology that allows gamers to control game functions with only their eyes is helping to open virtual worlds to people with severe motor disabilities. The gaming-with-gaze software, which uses gaze- and eye-tracking technology, has been



iPoint 3D

made publicly available for people with disabilities. For people suffering from conditions such as cerebral palsy, motor neuron disease or so-called locked-in syndromes, being able to move around and interact in a virtual environment will be a truly liberating experience.

Fingertip digitiser

With the tap of a single finger, computer users soon may be drawn deeper into the virtual world using a new device developed at the University at Buffalo's Virtual Reality Lab.

The Fingertip Digitizer, which users wear on the tip of their index finger, can transfer to the virtual world the meaning and intent of common hand gestures such as pointing, wagging the finger or tapping in the air, which can be used to direct the actions of an electronic device—much like a mouse directs the actions of a personal computer, but with greater precision. What's more, the Fingertip Digitizer can transfer to personal computers very precise information about the physical characteristics of an object—and can even sense the shape and size of a human gland or tumour—when the user taps, scratches, squeezes, strokes or glides a finger over the surface of the object.

The gesture-recognition function of this device has great potential for applications ranging from personal computing to medical diagnostics to computer games. The Fingertip

Digitizer will help bridge the gap between what a person knows and what a computer knows. With this device, a computer, cellphone or computer game could read human intentions more naturally. Eventually, the Fingertip Digitizer may be used as a high-end substitute for a mouse, keyboard or joystick.

The road ahead

Interactive electronics manufacturers are engaged in the fusion of various media and technologies including live video, film, moving image, graphics, lasers, electronics, optics, computers, software, sensors and microcontrollers. So a lot more interactive electronic devices with user-friendly and advanced features are expected to emerge in coming times.

Because human-machine interaction study involves a human and a machine in conjunction, it draws from supporting knowledge on both the machine and the human side. A great attention to human-machine interaction is important because poorly designed human-machine interfaces may lead to many unexpected problems. ●

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